

Geography Ch-15 (Advanced): India's Equatorial-Type Rainforests - Western Ghats & Northeast

Tier 2: Advanced | Source: D.R. Khullar + Majid Husain (Geography of India)
India-applied grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

India's Tropical Evergreen (Equatorial-Type) Forests: Western Ghats, NE & A&N Islands

GS-I (Physical Geography of India / Biogeography) Geography

■ NEWS TRIGGER

Advanced application (D.R. Khullar; Majid Husain). Though India lacks a true equatorial (Af) climate, its wettest, near-equatorial-type tropical evergreen rainforests occur where rainfall exceeds 200-250 cm - the windward Western Ghats, the NE hills and the Andaman & Nicobar Islands.

■ INTRO & ORIGIN

India's tropical wet evergreen forests mimic the equatorial rainforest in structure and richness. They occupy the heavy-rainfall (>200 cm), short-dry-season belts - the Malabar coast/Western Ghats, the NE states, and A&N - and are biodiversity hotspots central to the Ghats' and NE's ecological importance.

Koppen Amw (tropical monsoon, West Coast) approaches equatorial conditions on the windward Ghats (orographic rain > 250 cm). These evergreen forests are structurally similar to selva - multi-layered, dense hardwoods.

■ KEY DATA

Region	Forest type	Key species / note
Windward Western Ghats	Tropical wet evergreen	Rosewood, mahogany, ebony; >250 cm rain
NE hills (Assam, Meghalaya)	Tropical wet evergreen / semi-evergreen	Mawsynram-Cherrapunji: world's wettest
Andaman & Nicobar	Tropical evergreen + littoral	Insular rainforest, high endemism
Foothills of E. Himalaya	Semi-evergreen	Transition to monsoon forest

■ STATIC THEORY

- **Tropical wet evergreen forests (India):** found where annual rainfall > 200-250 cm and the dry season is short; trees are tall (up to 45-60 m), evergreen, broad-leaved hardwoods forming a closed, multi-storeyed canopy - structurally the Indian analogue of equatorial selva.
- **Distribution:** windward Western Ghats (Malabar), the NE states (Assam, Meghalaya, Arunachal), and the Andaman & Nicobar Islands.
- **Climatic controls:** heavy orographic monsoon rain (SW monsoon striking the Ghats/NE) creates equatorial-like moisture surplus; Mawsynram-Cherrapunji (Meghalaya) are the wettest places on earth.
- **Ecological value:** the Western Ghats and NE India are among the world's biodiversity hotspots; these forests hold high endemism and are UNESCO/hotspot-recognised.
- **Shifting cultivation (jhum):** in the NE, the traditional slash-and-burn 'jhum' cultivation mirrors equatorial shifting agriculture - now under pressure as fallow cycles shorten (D.R. Khullar).

■ MUST-KNOW FACTS

1. India has no true Af equatorial climate; its equivalent is tropical wet evergreen forest (>200-250 cm rain).
2. Located: windward Western Ghats (Malabar), NE India, and Andaman & Nicobar Islands.

■ UPSC TRAPS

- ✗ India has a true equatorial (Af) climate in the south.
- ✓ India has NO true Af climate; the closest is tropical monsoon (Amw) on the windward Ghats, supporting evergreen forests.

3. Mawsynram & Cherrapunji (Meghalaya) = wettest places on earth (orographic monsoon rain).

4. Western Ghats + NE India = global biodiversity hotspots with very high endemism.

5. Jhum (NE shifting cultivation) is the Indian analogue of equatorial slash-and-burn.

✗ The leeward (eastern) side of the Western Ghats has evergreen rainforest.

✓ Evergreen forest is on the WINDWARD (western) slope; the leeward rain-shadow side is dry deciduous/scrub.

■ MAINS ANGLE

Explain why India's tropical wet evergreen forests, though not under a true equatorial climate, replicate equatorial rainforest structure - and why the Western Ghats and NE are biodiversity hotspots needing protection.

■ STUDY LINK

Geography -> Natural Vegetation of India -> Tropical Wet Evergreen; connects to Western Ghats hotspot & jhum cultivation

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: Western Ghats Eco-Sensitive Zone & Shrinking Jhum Cycles

GS-I (Geography of India) + GS-III (Environment / Conservation)
Geography - Current Affairs

■ NEWS TRIGGER

2024-26: fresh MoEFCC draft notifications on the Western Ghats Ecologically Sensitive Area (ESA) - reviving the Gadgil/Kasturirangan debate - and studies flagging that shortened jhum (shifting cultivation) fallow cycles in the NE are degrading India's evergreen rainforests.

■ INTRO & ORIGIN

India's equatorial-type rainforests are under a twin squeeze: the Western Ghats ESA notification struggle (how much to protect vs develop) and the ecological breakdown of jhum in the NE as fallow periods shrink from 15-20 years to 3-5 - both directly relevant to conserving tropical wet evergreen forest.

Gadgil Committee (2011) recommended ~64% of the Ghats as ESA; Kasturirangan Committee (2013) reduced it to ~37% (only 'natural' landscape). Successive draft ESA notifications since remain contested by states.

■ KEY DATA

Issue	Detail	Significance
Western Ghats ESA	Gadgil ~64% vs Kasturirangan ~37%	Balancing conservation vs development
Draft ESA notifications	Repeated MoEFCC drafts 2014-2026	States resist; hotspot still unprotected
Jhum fallow collapse	15-20 yr -> 3-5 yr cycles (NE)	Soil exhaustion, forest degradation

■ STATIC THEORY

- **Ecologically Sensitive Area (ESA):** a zone notified under the Environment (Protection) Act, 1986 restricting mining, quarrying, red-category industry and large construction to protect fragile ecosystems.
- **Gadgil vs Kasturirangan:** the core policy tension - a maximalist ecological approach (Gadgil, ~64%) vs a development-accommodating approach (Kasturirangan, ~37%, natural landscape only).
- **Jhum degradation:** traditional shifting cultivation was sustainable when fallow (regrowth) periods were long; population pressure has shortened cycles, preventing forest recovery and causing erosion.

■ MUST-KNOW FACTS

■ UPSC TRAPS

1. Gadgil Committee (2011): ~64% of Western Ghats as ESA; Kasturirangan (2013): ~37%. 2. ESA notified under the Environment (Protection) Act, 1986. 3. Western Ghats is a UNESCO World Heritage Site and a global biodiversity hotspot. 4. NE jhum fallow cycles have collapsed from ~15-20 years to ~3-5 years, degrading evergreen forest. 5. Repeated draft ESA notifications remain contested by Kerala, Karnataka, etc.	✗ The Kasturirangan report recommended a larger ESA than Gadgil. ✓ Kasturirangan REDUCED the ESA (~37%, natural landscape only) compared with Gadgil's ~64%. ✗ Jhum cultivation is inherently destructive. ✓ It was sustainable with long fallow cycles; degradation arises when population pressure shortens the fallow period.
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■ MAINS ANGLE

Discuss the Gadgil-Kasturirangan debate on the Western Ghats ESA and the sustainability crisis of jhum cultivation as two fronts in conserving India's equatorial-type rainforests.

■ STUDY LINK

Geography -> Western Ghats/NE forests + GS-III conservation (ESA, EPA 1986, jhum); connect to biodiversity hotspots

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